Onurcan Bektas · PhD candidate · LMU München · onurcan.bektas@lmu.de

KRAFT: Discovering how to preserve muscle strength

Uncovering the mechanisms of muscle repair to identify therapeutic targets for preserving muscle strength during ageing

As we age, our muscles lose their ability to repair themselves and that leads to muscle decline [1, 2]. Muscle decline is associated with 88% higher rate of death [3], affects around 20M people [4] and could cost around €40B every year in EU28 countries according to our estimates [5]. KRAFT will test whether preserving muscle repair can prevent age-related loss of muscle strength.

There are currently **no approved drugs** specifically designed to prevent or reverse age-related loss of muscle function [6]. Although companies are developing potential treatments [6], demonstrating their effectiveness is particularly difficult: muscle function declines gradually with age [7], requiring longer and more expensive clinical trials [8, 9]. This makes selecting the right therapeutic targets early in drug development especially important, as better target selection can reduce both development costs and the risk of failure [10, 11]. Yet we lack reliable models to systematically identify which biological mechanisms should be targeted [12, 13]. KRAFT will develop **new causal machine learning methods from first principles** to uncover how muscle repair works and predict which interventions could preserve it as we age. By testing these predictions experimentally, we aim to identify therapeutic targets that could prevent age-related muscle decline.

Recent evidence suggests that learning biological mechanisms from multi-modal measurements is possible [14]. For example, in a 2024 study, researchers combined measurements of gene activity, chromatin accessibility, and spatial gene expression to understand why human skin heals poorly years after radiotherapy. Gene activity measurements alone did not reveal the underlying problem: a gene called THBS1 showed no significant change in uninjured skin. However, chromatin measurements revealed that the gene had been left in a primed state, causing excessive activation following injury and impairing tissue repair. Researchers subsequently demonstrated that blocking THBS1 improved wound healing in human skin samples. This provides proof of principle that multi-modal measurements can reveal mechanisms missed by unimodal approaches and lead to experimentally supported therapeutic targets.

Because muscle repair depends both on molecular processes within individual cells and on interactions between different cell types [1, 2], KRAFT will specifically focus on **multi-modal, multi-cellular approaches.**

Recent technological breakthroughs make this approach **feasible**. Between 2016 and 2025, computing power per dollar increased 17-fold [15]; sequencing costs fell 3-fold between 2020 and 2022 [16, 17]; and spatial transcriptomics achieved 27-fold higher resolution between 2019 and 2024 [18, 19]. These advances are beginning to translate into scientific breakthroughs: in 2025, researchers demonstrated that biological mechanisms can be learned from data [20] and that studying highly regenerative species can reveal repair targets that also work in mammals [21]. In 2026, new technologies further expanded our ability to measure epigenetic and chromatin states together [22]. Meanwhile, our ability to intervene is also advancing: the FDA approved the first gene-editing therapy in 2023 [23], and **30 in vivo editing therapies are progressing through clinical trials as of 2026** [24], including applications targeting skeletal muscle [25]. Together, these advances **create an opportunity to move from understanding** muscle repair to identifying and testing interventions that could **preserve it** during aging.

KRAFT will run as a **three-year, €60M Challenge**, because we know that success means maintaining muscle strength as we age, but we do not know which approach will work best. In the first 18 months (€35M), we will bring together eight competing teams and provide them with existing biological data to develop models of muscle repair. After six months, we will evaluate their models against newly generated, high-quality experimental data that the teams have not seen and select four teams based on their ability to predict experimental outcomes. Between months 6 and 18, we will progressively test these predictions in vitro, ex vivo and in vivo, measuring their effects on muscle repair and function. In parallel, we will test engaging with these targets using existing drugs. At month 18, we will select two teams based on functional outcomes of the targets, promising therapeutic leads and the accessibility of the targets. In the remaining 18 months (€25M), we will optimise these leads and select development candidates by month 24. We will engage regulators during candidate development and aim to obtain feedback by month 24 through FDA pre-IND meetings or European scientific advice. This feedback will inform the definitive preclinical safety studies. Studies of drug exposure and biological activity (PK/PD), efficacy and preliminary tolerability will begin during candidate development, while manufacturing development and preparation for clinical testing will proceed in parallel. Efficacy studies will assess whether improved repair translates into preserved muscle strength over time. The second-stage budget comprises €20M in planned funding and €5M to be released conditionally following review of the most promising candidates' progress towards clinical testing. By month 36, we aim to have experimentally validated therapeutic

targets and **at least one therapeutic candidate ready for an IND application in the US or a CTA in the EU**, supported by the required preclinical efficacy and safety data.

KRAFT starts with muscle. Its ultimate ambition is a programmable, self-repairing body. Imagine muscles that stay powerful into old age, damaged organs that rebuild themselves, and cells guided back to health before their malfunction becomes disease. In this future, medicine would shift from a reactive practice to a preventative one. KRAFT takes the first step towards that future by learning what drives muscle to deteriorate and how to shift that balance towards repair. The long-term goal is to extend that control throughout the body.

REFERENCES

[1] Walter LD, Orton JL, Ntekas I, et al. Transcriptomic analysis of skeletal muscle regeneration across mouse lifespan identifies altered stem cell states. Nature Aging. 2024;4:1862-1881. doi:10.1038/s43587-024-00756-3. https://doi.org/10.1038/s43587-024-00756-3

[2] Kedlian VR, Wang Y, Liu T, et al. Human skeletal muscle aging atlas. Nature Aging. 2024;4:727-744. doi:10.1038/s43587-024-00613-3. https://doi.org/10.1038/s43587-024-00613-3

[3] Xu J, Wan CS, Ktoris K, Reijnierse EM, Maier AB. Sarcopenia Is Associated with Mortality in Adults: A Systematic Review and Meta-Analysis. Gerontology. 2022;68(4):361-376. doi:10.1159/000517099. https://doi.org/10.1159/000517099

[4] Ethgen O, Beaudart C, Buckinx F, Bruyere O, Reginster JY. The Future Prevalence of Sarcopenia in Europe: A Claim for Public Health Action. Calcified Tissue International. 2017;100:229-234. doi:10.1007/s00223-016-0220-9. https://doi.org/10.1007/s00223-016-0220-9

[5] Pinedo-Villanueva R, Westbury LD, Syddall HE, et al. Health Care Costs Associated With Muscle Weakness: A UK Population-Based Estimate. Calcified Tissue International. 2019;104:137-144. doi:10.1007/s00223-018-0478-1. https://doi.org/10.1007/s00223-018-0478-1

[6] Rejuvenate Biomed. Rejuvenate Biomed Completes Recruitment in Phase 2 Trial

Assessing First-in-Class Multi-Pathway Therapeutic RJx-01 in COPD-Induced Sarcopenia. Company press release. 1 June 2026. https://www.rejuvenatebiomed.com/en/news/rejuvenate-biomed-completes-recruitment-in-phase-2-trial-assessing-first-in-class-multi-pathway-therapeutic-rjx-01-in-copd-induced-sarcopenia

[7] Goodpaster BH, Park SW, Harris TB, et al. The Loss of Skeletal Muscle Strength, Mass, and Quality in Older Adults: The Health, Aging and Body Composition Study. Journals of Gerontology Series A: Biological Sciences and Medical Sciences. 2006;61(10):1059-1064. doi:10.1093/gerona/61.10.1059.

[8] Reginster JY, Beaudart C, Al-Daghri N, et al. Update on the ESCEO recommendation for the conduct of clinical trials for drugs aiming at the treatment of sarcopenia in older adults. Aging Clinical and Experimental Research. 2021;33(1):3-17. Published online 31 July 2020. doi:10.1007/s40520-020-01663-4. https://doi.org/10.1007/s40520-020-01663-4

[9] Sertkaya A, Beleche T, Jessup A, Sommers BD. Costs of Drug Development and Research and Development Intensity in the US, 2000-2018. JAMA Network Open. 2024;7(6):e2415445. doi:10.1001/jamanetworkopen.2024.15445. https://doi.org/10.1001/jamanetworkopen.2024.15445 https://aspe.hhs.gov/reports/drug-development-rd-intensity

[10] Minikel EV, Painter JL, Dong CC, Nelson MR. Refining the impact of genetic evidence on clinical success. Nature. 2024;629:624-629. doi:10.1038/s41586-024-07316-0. https://doi.org/10.1038/s41586-024-07316-0

[11] Morgan P, Brown DG, Lennard S, et al. Impact of a five-dimensional framework on R&D productivity at AstraZeneca. Nature Reviews Drug Discovery. 2018;17:167-181. doi:10.1038/nrd.2017.244. https://doi.org/10.1038/nrd.2017.244

[12] Ahlmann-Eltze C, Huber W, Anders S. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. Nature Methods. 2025;22:1657-1661. doi:10.1038/s41592-025-02772-6. https://doi.org/10.1038/s41592-025-02772-6

[13] Wei Z, Wang Y, Gao Y, et al. Benchmarking algorithms for generalizable single-cell perturbation response prediction. Nature Methods. 2026;23:451-464. Published online

11 December 2025. doi:10.1038/s41592-025-02980-0.
https://doi.org/10.1038/s41592-025-02980-0

[14] Bian X, Piipponen M, Liu Z, et al. Epigenetic memory of radiotherapy in dermal fibroblasts impairs wound repair capacity in cancer survivors. Nature Communications. 2024;15:9286. doi:10.1038/s41467-024-53295-1.
https://doi.org/10.1038/s41467-024-53295-1

[15] Stanford Institute for Human-Centered Artificial Intelligence (HAI). The 2025 AI Index Report. Stanford University; 2025

[16] Illumina. Illumina Announces the NovaSeq 6000 v1.5 Reagent Kit Unlocking Deeper Discoveries with the $600 Genome. Company press release. 2020.
https://www.illumina.com/company/news-center/press-releases/2020/9c48adf5-5b78-4e18-8116-3c7c8b3ad79f.html

[17] Illumina. Illumina's revolutionary NovaSeq X exceeds 200th order milestone in first quarter 2023. Company press release. 4 April 2023.
https://www.illumina.com/content/illumina-marketing/amr/en_US/company/news-center/press-releases/2023/4b48580b-42d4-419b-8512-5adcbb069836.html

[18] 10x Genomics. Visium HD Spatial Gene Expression Performance. Technical Note CG000686, Revision A. 2024.
https://cdn.10xgenomics.com/image/upload/v1722285090/support/documents/CG000686_VisiumHDSpatialGeneExpressionPerformance_TechnicalNote_RevA.pdf

[19] 10x Genomics. 10x Genomics Begins Shipments of Visium Spatial Gene Expression Solution. Company press release, distributed by GlobeNewswire and reproduced by Nasdaq. 26 November 2019.
https://www.nasdaq.com/press-release/10x-genomics-begins-shipments-of-visium-spatial-gene-expression-solution-2019-11-26

[20] Brown BC, Tokolyi A, Morris JA, Lappalainen T, Knowles DA. Large-scale causal discovery using interventional data sheds light on gene network structure in k562 cells. Nature Communications. 2025;16:9628. doi:10.1038/s41467-025-64353-7.
https://doi.org/10.1038/s41467-025-64353-7

[21] Lin W, Jia X, Shi X, et al. Reactivation of mammalian regeneration by turning on an evolutionarily disabled genetic switch. Science. 2025;388(6754):eadp0176.
doi:10.1126/science.adp0176.

https://doi.org/10.1126/science.adp0176

[22] Araujo Tavares RdC, Dhir S, He X, et al. Sequencing DNA methylation and hydroxymethylation at co-occurring chromatin features. Nature Communications. 2026;17:2591. doi:10.1038/s41467-026-69429-6.
https://doi.org/10.1038/s41467-026-69429-6

[23] U.S. Food and Drug Administration. FDA Approves First Gene Therapies to Treat Patients with Sickle Cell Disease. FDA news release. 8 December 2023.
https://www.fda.gov/news-events/press-announcements/fda-approves-first-gene-therapies-treat-patients-sickle-cell-disease

[24] Lummerstorfer M, Lächelt U. Non-viral CRISPR carriers: transient delivery with lasting effects. Drug Delivery. 2026;33(1):2614125.
https://doi.org/10.1080/10717544.2026.2614125

[25] Precision BioSciences. Precision BioSciences Commences Dosing in Phase 1/2 FUNCTION-DMD Study. Company press release. 24 August 2026. Trial registration: NCT07429240.
https://investor.precisionbiosciences.com/news-releases/news-release-details/precision-biosciences-commences-dosing-phase-12-function-dmd
https://clinicaltrials.gov/study/NCT07429240